

BITS, PILANI – K. K. BIRLA GOA CAMPUS

Theory of Computation

(CS F51)

Lecture CFG - 4



Ambiguity: Computational Problems

Problem1: Given CFG G , find CFG G' such that $L(G) = L(G')$ and G' is unambiguous.

There is *no* algorithm that can solve the above problem!

Problem2: Given CFG G , determine if G is ambiguous.

The problem is *undecidable*.

Problem3: Is it the case that for every CFG G , there is a grammar G' such that $L(G) = L(G')$ and G' is unambiguous, even if G' cannot be constructed algorithmically?

No! There are context-free languages L such that every grammar for L is ambiguous.

Inherently ambiguous language

A context-free language L is said to be *inherently ambiguous* if *all* its grammars are *ambiguous*

Fortunately, programming languages are never *inherently ambiguous*.

Example of a language that can be proved *inherently ambiguous*.

$$L = \{a^n b^n c^m d^m \mid n \geq 1, m \geq 1\} \cup \{a^n b^m c^m d^n \mid n \geq 1, m \geq 1\}$$

$$S \rightarrow AB \mid C$$

$$A \rightarrow aAb \mid ab \qquad B \rightarrow cBd \mid cd$$

$$C \rightarrow aCd \mid aDd \qquad D \rightarrow bDc \mid bc$$

Properties of CFL

Theorem :

The class of context free languages is closed under Union, Concatenation and Kleene Star.

Proof: Let L_1 and L_2 be cfl's generated by cfg's $G_1 = (V_1, \Sigma_1, R_1, S_1)$ & $G_2 = (V_2, \Sigma_2, R_2, S_2)$ respectively.

Union

Let S be a new symbol and let

$G = (V_1 \cup V_2 \cup \{S\}, \Sigma_1 \cup \Sigma_2, R, S)$ where

$R = R_1 \cup R_2 \cup \{S \rightarrow S_1, S \rightarrow S_2\}$

We can show that $L(G) = L(G_1) \cup L(G_2)$

Properties of CFL

Concatenation

$G = (V_1 \cup V_2 \cup \{S\}, \Sigma_1 \cup \Sigma_2, R, S)$ where

$$R = R_1 \cup R_2 \cup \{S \rightarrow S_1 S_2\}$$

We can show that $L(G) = L(G_1)L(G_2)$

Kleene Star

$G = (V_1 \cup \{S\}, \Sigma_1, R, S)$ where

$$R = R_1 \cup \{S \rightarrow e, S \rightarrow SS_1\}$$

We can show that $L(G) = L(G_1)^*$

Problems on CFG

Problems on CFG

1. $L = \{a^m b^n \mid m \geq n\}$
2. $L = \{a^{2^n} b^n \mid n \geq 0\}$
3. $L = \{a^m b^n \mid m \geq 2n\}$
4. $L = \{a^m b^n \mid m \leq 2n\}$
5. $L = \{a^m b^n \mid n \leq 2m\}$
6. Find cfg for language in which $n_a(w) = n_b(w)$
Derive aababb
7. Find cfg for language in which $n_a(w) \neq n_b(w)$
8. $L = \{a^m b^n c^p d^q \mid m + n = p + q\}$